Healthcare Presentation

AI-Driven

PHARMACEUTICAL COLD-CHAIN MONITORING
& DYNAMIC FEFO INFORMATION SYSTEM

The Paradigm Shift: From Static Dates to Dynamic Quality Risk

	Traditional FEFO	**Dynamic FEFO**
Trigger Points	Fixed label dates	• Arrhenius-adjusted dynamic dates based on true thermal history.
Data Logging	Manual logging and discrete spot-calls.	• Continuous IoT telemetry (capturing temperature and humidity hourly).
Incident Response	Reactive (alerts trigger after damage occurs).	• Predictive (XGBoost models forecast excursions 3-6 hours in advance).
Dispatch Logic	First-Expired, First-Out (Chronological).	• Highest-Risk, First-Out (Cumulative Exposure).

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THE VALUE DELIVERED: REMOVING GUESSWORK FROM COLD CHAIN LOGISTICS

MONITOR Bridging biological realities via continuous Arrhenius kinetics, moving past static label dates.

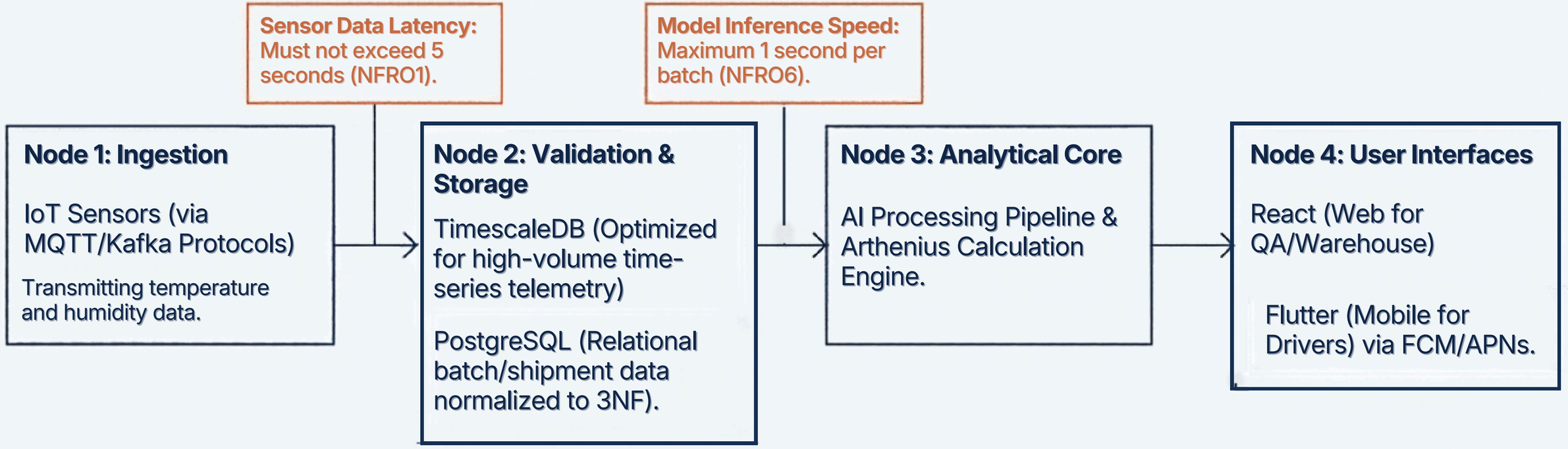
ASSESS Anticipating failures via XGBoost predictive intelligence before they impact product integrity.

PROTECT Operationalizing risk through Dynamic FEFO, governed absolutely by human QA authority and rigorous Policy Regression.

Protecting product integrity by mathematically anticipating risk, while keeping the human in control.

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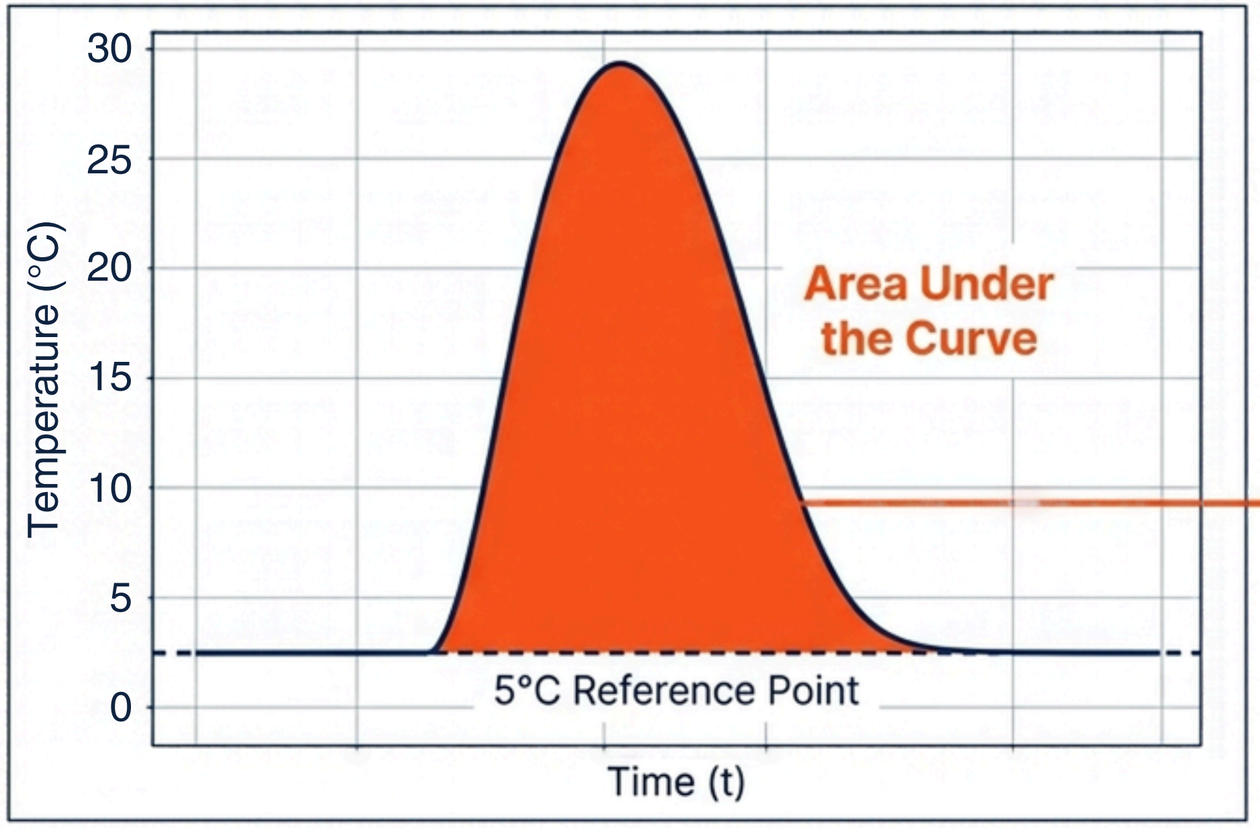
System Architecture: Telemetry & Data Processing Pipeline



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The Biological Reality: Quantifying Decay via Arrhenius Kinetics

Key Insight: Pharmaceutical stability is biochemical, not temporal. Decay only accumulates when the **Relative Degradation Rate is > 1** (degrading faster than the 5°C reference).



Equations & Parameters

- Reaction Rate: $k(T) = A \times \exp\left(-\frac{Ea}{RT}\right)$
- Prototype Parameters: E_a = 51044.8, T_{ref} = 5°C
- **Outcome Metric: Dynamic Expiry = Label Expiry – Days Lost**

Context Note: The system continuously integrates historical temperature fluctuations to calculate **cumulative biochemical decay in real-time** without altering original data files.

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Predictive Intelligence: Distinguishing ML Capabilities

Panel 1: Contextual Anomaly Detection

Mechanism: Evaluates recent temperature/humidity time-series windows to detect sudden environmental shifts (e.g., prolonged door opening during loading).

Output Metric: Anomaly Score: 0 to 100.

Status: Prototype uses static anomaly labels strictly for historical reference.



Panel 2: XGBoost Event Risk Forecasting

Mechanism: Projects the probability of a severe excursion (temperature dropping outside 2-8°C for >3 hours) within the next 6, 12, or 24 hours.

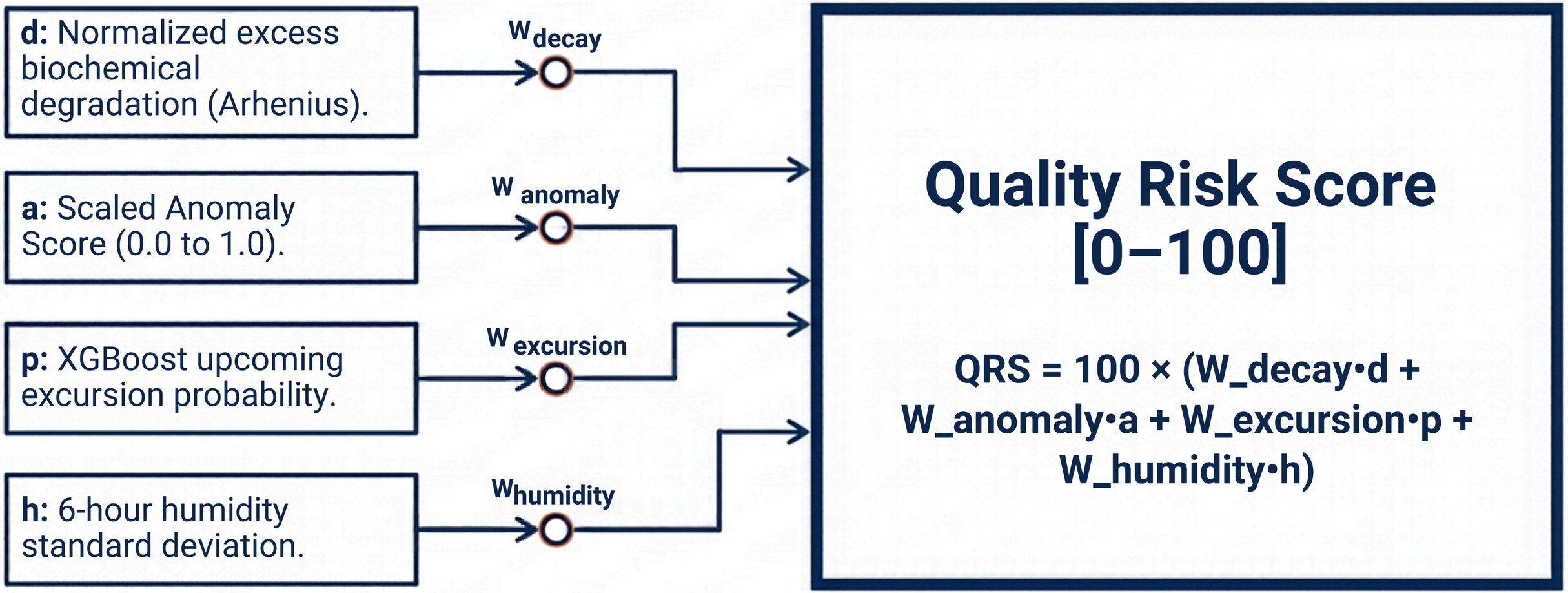
Condition: Only evaluates batches currently within safe bounds. If currently out-of-bounds, mechanism displays “N/A”.

Outputs: Calibrated Probability (p_{cal}), Alert Threshold (e.g., Precision50), and Alert Status.



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Synthesizing the Quality Risk Score (QRS)



Constraint Rule: W_de*cay* + W*a*noma*iy* + W*excr*sion + Whumidity = 1

Audit Principle: Risk policies dictate exact weights, ensuring QRS remains permanently auditable against the policy version active at runtime.

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Operationalizing Risk: Dynamic FEFO Prioritization

Sorting Logic:

- Primary Sort: Descending by Quality Risk Score (prioritizing batches with the highest cumulative exposure).
- Secondary Sort: If QRS is identical, sort by lowest Remaining Days.

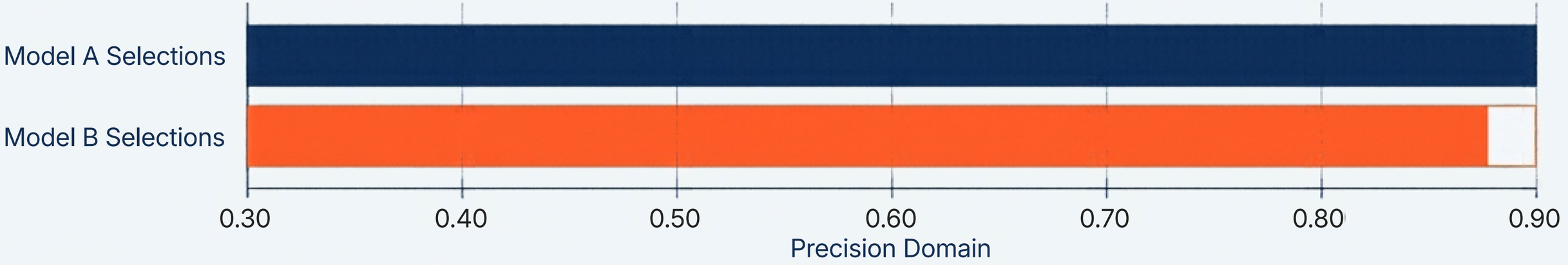
Warehouse Dispatch Board - Eligible Pool					
Batch ID	**Label Expiry**	**Dynamic Expiry**	**QRS**	**Primary Risk Contributor**	**Priority Rank**
Batch B-992	365 Days	142 Days Remaining	88/100	Temp Excursion Forecast	1
Batch A-410	365 Days	180 Days Remaining	64/100	Arrhenius Decay	2
Batch A-411	365 Days	205 Days Remaining	64/100	Arrhenius Decay	3
Batch C-887	365 Days	310 Days Remaining	12/100	None	4

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Algorithmic Governance: Policy Regression Diagnosis

The Challenge: Tracking general data drift is insufficient. The system must mathematically guarantee that a new model version does not unintentionally alter the exact selected population of batches flagged for intervention.

Demo Execution Parameters: Precision Domain tested from 0.30 to 0.90 (Inclusive ≥).



Outcome: Equivalent

Models select the exact same Batch IDs across the domain.

Outcome: Divergent

Models select different populations. System flags the exact Witness ID, calculates the PSC (Precision Selection Contract) length, and generates a SHA-256 JSON certificate for audit.

Immutable Audit Log Output


```
{"witness_id": "89f7a2b1c4e6d3",
 "timestamp": "2024-08-15T14:22:10Z",
 "divergence_event": true,
 "psc length": 0.10,
 "sha256-certificate":
 "e3b8c44298fc1c149afbfac8996fb9242
 7ae41e4649b934ca495991b7852b855"
```


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Implementation Status & Scalability Architecture

Test Scale Context: Prototype rigorously validated on a dataset of 1,500 batches / 416,688 records.

	Current Prototype	Target Design
Arhhenius Parameters	Prototype utilizes fixed parameters (calculated at current timestamp).	Target Design integrates product-specific metrics calculated over continuous history.
Predictive Scope	Prototype forecasts probability of >3h out-of-bounds events.	Target Design predicts exact temperatures at specific 3h/6h intervals.
System Deployment	Prototype currently operates via a local Python launcher (limited to 2,000 observations per run).	Target Design features fully automated pipeline deployment with seamless regression.

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Role-Based Access: 3 Interfaces, 1 Unified Identity

1 Unified Identity



QA Manager
(ColdChain IQ Web)

- Permissions: Full access to system alerts, historical simulation, and final approval of warehouse dispatch.



Warehouse Staff
(ColdChain IQ Web)

- Permissions: View alerts only. Dispatch board is **strictly locked** until QA grants approval. No AI knowledge required.

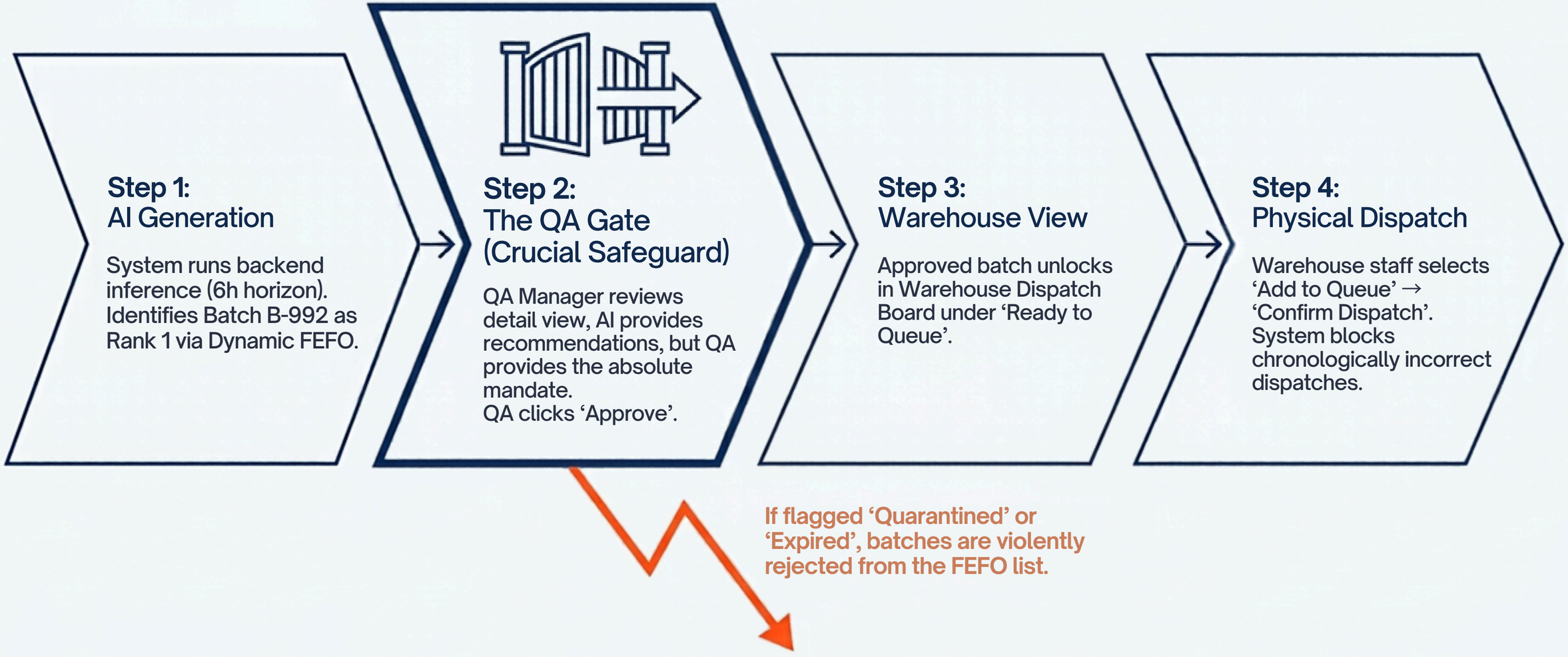


Driver
(ColdRoute Mobile)

- Permissions: Receive **pre-transit early warnings** (e.g., check refrigeration unit). **Submit incident action reports** via mobile UI. No approval permissions.

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The Closed-Loop Workflow & QA Eligibility Gate



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THANK YOU!